

PATRICK KIPROP ROTICH

ICT Support | IT Operations | Software & Systems Support | Network & Security

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PROFESSIONAL PROFILE

Computer Science graduate (2026) with hands-on experience across ICT support, systems deployment, networking, and software development. Gained enterprise ICT experience during industrial attachment at Kenya Revenue Authority, supporting users, installing and configuring operating systems and applications, and resolving connectivity and hardware/software issues.

Complementary practical experience in backend development (Python, FastAPI), databases, Linux, Docker, and network security. Comfortable working across infrastructure, technical support, and application development roles.

Seeking opportunities in **IT support, systems administration, software development, technical operations, or cybersecurity**, where I can contribute to reliable systems and user-focused technology solutions.

CORE COMPETENCIES

Networking	TCP/IP, LAN/WAN, connectivity troubleshooting, structured cabling, packet analysis, port scanning
IT Support	Hardware/software troubleshooting, OS installation, desktop deployment, system configuration, user support
Network Security	Intrusion detection, network monitoring, anomaly detection, packet analysis, SSH brute-force detection
Systems	Windows/Linux, server assistance, Docker, Docker Compose, Git/GitHub
Development	Python, C++, SQL, FastAPI, REST APIs
IoT and Embedded	MQTT, ESP8266, sensor integration, data publishing and monitoring

PROFESSIONAL EXPERIENCE

ICT Intern (Industrial Attachment) — Kenya Revenue Authority (KRA), ICT Department | 2025–2026

- Provided end-user ICT support, troubleshooting hardware and software issues and assisting users with technical problems.
- Installed and configured Windows/Linux operating systems and business applications for users and workstations.
- Assisted with structured cabling and basic network troubleshooting, including TCP/IP connectivity issues.
- Performed desktop deployment and system configuration for new and existing users.
- Assisted with server-related monitoring and basic maintenance under senior ICT supervision.
- Documented technical issues, resolutions and user-support feedback for follow-up and service improvement.
- Applied network fundamentals, hardware configuration, ICT service-desk practices and client communication in a professional environment.

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EDUCATION

Bachelor of Science in Computer Science University of Embu

Relevant coursework: Computer Networks, Distributed Systems, Information Systems, Machine Learning, ICT Project Management.

TECHNICAL PROJECTS

Software & Backend Development

IoT Device Management REST API

- Developed a FastAPI backend for registering, monitoring, and controlling IoT devices remotely.
- Implemented SQLite device registry and telemetry storage with device health and last-seen monitoring.
- Demonstrates REST API design, database modelling, CRUD operations, and backend architecture.

Point of Sale (POS) System — Full Stack Development

- Developed a full-stack POS application with backend and frontend components for sales processing and business logic.
- Implemented user-interface modules and system functionality demonstrating practical application development.

- GitHub: github.com/Rotichpatrick792/pos-backend | github.com/Rotichpatrick792/pos-frontend

Afya Bora Health System

- Developed a healthcare-related system focused on access to basic services and structured information management.
- Applied user-centric system design and data-handling concepts.
- GitHub: github.com/Rotichpatrick792/afya-bora

Job Connector AI — Career Matching System

- Built a system for connecting users with job opportunities using structured matching logic.
- Focused on backend logic and data-driven decision-making workflows.
- GitHub: github.com/Rotichpatrick792/job-connector

Fault Detection System

- Designed a monitoring system for identifying device or system faults using logical condition monitoring.
- Implemented anomaly identification and alert triggering based on predefined system behaviours.
- GitHub: github.com/Rotichpatrick792/fault-detector

Networking, Infrastructure & IoT

Network Intrusion Detection System

- Built a real-time packet sniffer to analyze network traffic and identify port scans and SSH brute-force activity.
- Used Scapy for packet capture and analysis and tracking logic for connection attempts.
- Implemented automated IP blocking and alert generation for suspicious activity.
- Demonstrates practical understanding of network security, packet analysis, anomaly detection, and real-time monitoring.

Cross-Platform Firewall System

- Developed a rule-based firewall system focused on traffic filtering and network protection concepts.

- Implemented rule-based filtering logic to simulate cybersecurity enforcement mechanisms.
- Demonstrates understanding of network security principles relevant to enterprise and IoT environments.
- GitHub: github.com/Rotichpatrick792/cross-platform-firewall

MQTT Sensor Network with ESP8266

- Programmed an ESP8266 microcontroller with a DHT11 temperature/humidity sensor to collect environmental data.
- Published sensor data to a cloud MQTT broker using the publish/subscribe communication model.
- Developed a Python subscriber for data ingestion and storage.
- Demonstrates embedded programming, hardware integration, MQTT networking, sensor calibration, and serial communication.

Dockerized IoT Platform

- Containerized an IoT platform using Docker Compose with Mosquitto MQTT, InfluxDB, Grafana, and a Python consumer.
- Configured container networking and volume management across multiple services.
- Created a one-command deployment workflow using docker-compose up.
- Demonstrates containerization, infrastructure deployment, networking, and DevOps practices.

TECHNICAL SKILLS — DETAILED

Networking: TCP/IP, LAN/WAN, connectivity troubleshooting, structured cabling, packet analysis, port scanning

Network Security: Intrusion detection, Scapy, network monitoring, anomaly detection, firewall concepts

Operating Systems: Windows, Linux, system installation and configuration

Programming: Python, C++ (Arduino), SQL

IoT & Embedded: ESP8266, Arduino IDE, DHT11, MQTT, sensor integration

Backend & Databases: FastAPI, REST APIs, SQLite, CRUD operations

Infrastructure & DevOps: Docker, Docker Compose, container networking, volume management

Tools: Git/GitHub, Cisco Packet Tracer, Kali Linux, VS Code

PROFESSIONAL STRENGTHS

- Analytical troubleshooting and structured problem-solving.
- User-focused approach to ICT support and technical issue resolution.
- Practical understanding of networking and network security concepts.
- Ability to document technical issues, procedures and solutions.
- Hands-on approach to learning new technologies and working with hardware and software.
- Comfortable working independently while escalating complex technical issues when required.

CERTIFICATIONS & TRAINING

Currently Pursuing: Cisco Introduction to IoT

RELEVANT ACADEMIC COURSEWORK

- Computer Networks
- Distributed Systems
- Information Systems
- Machine Learning
- ICT Project Management

PERSONAL INFORMATION

Nationality	Kenyan
Languages	English (Fluent), Kiswahili (Fluent)
Availability	Available full time
Work Authorization	Kenyan citizen
Driving Licence	Yes

REFERENCES

Steve Gitau — KRA Supervisor | 0726321415

Dr Faith — University of Embu Lecturer | 0737766052

DECLARATION

I certify that the information provided in this curriculum vitae is accurate to the best of my knowledge.